I AM CYBER SOLDIER

Cybersecurity Fundamentals – Basic Course

30-Hours (1.5 month) Beginner-Friendly | Theory + Hands-On Labs | Free Tools

LEARN → PRACTICE → INVESTIGATE → SECURE

Course Overview

This independent professional training program introduces learners to core cybersecurity concepts and gives them practical exposure to networking, security fundamentals, vulnerability assessment, authentication, access control, web security, cryptography, Linux security tools, SIEM investigation, AI-assisted security workflows, and real-world security scenarios.

The course is designed for beginners and uses free/open-source software and free learning/lab options where practical. Microsoft Sentinel and Splunk exercises should be performed only where a currently available free/trial entitlement exists; otherwise, the course provides local/synthetic alternatives.

Total Duration	30 Hours
Theory	15 Hours
Practical / Labs	15 Hours
Level	Beginner / Foundation
Delivery	Instructor-led + Hands-on
Primary Lab Model	Local isolated lab + free/trial SIEM options

Learning Outcomes

- Explain fundamental cybersecurity terminology, objectives, risks, threats, vulnerabilities and attacks.
- Understand basic TCP/IP networking and identify common network-security controls.
- Apply the CIA Triad to practical security scenarios.
- Perform basic vulnerability discovery and interpret findings in an authorized lab.
- Explain password security, MFA, authentication and authorization concepts.
- Understand access-control models and least privilege.
- Recognize common web-application security weaknesses and test a deliberately vulnerable local application.
- Explain hashing, symmetric encryption, asymmetric encryption, digital signatures and TLS at a foundational level.
- Use basic Linux commands and security tools such as Nmap and Wireshark.
- Perform introductory investigations in Splunk and Microsoft Sentinel using sample/synthetic data or eligible free/trial environments.
- Use AI responsibly for security learning, analysis assistance and defensive documentation.
- Investigate realistic security scenarios and produce a basic incident report.
- Complete a beginner cybersecurity capstone project.

Free / No-Cost Lab Stack

- Kali Linux – free distribution for security learning and auditing; official documentation recommends obtaining images from official sources and verifying them. □cite□turn0search3□turn0search7□
- Nmap – free network discovery and security auditing tool. □cite□turn0search0□turn0search2□

- Wireshark – free/open-source packet-analysis platform.
- OWASP ZAP – free web-application security testing tool suitable for beginners. □cite□turn0search10□
- OWASP Juice Shop / WebGoat – deliberately vulnerable applications for local training.
- VirtualBox or another available local hypervisor – use isolated VMs where supported by the host setup.
- Python – scripting and automation exercises.
- CyberChef – browser-based data transformation and encoding/decoding practice.
- Splunk – use an eligible free/trial or local training option; if unavailable, use provided synthetic logs and a local log-analysis alternative.
- Microsoft Sentinel – use the current Microsoft free-trial allowance where eligible; Microsoft currently documents a 31-day Sentinel free trial with the first 10 GB/day of Analytics logs free, while extra capabilities can still incur charges. □cite□turn0search9□
- Ollama or another locally runnable AI model – optional for AI-assisted exercises when the learner's hardware supports it.

Recommended Lab Architecture

- Host PC: Windows/Linux/macOS with at least 8 GB RAM recommended; 16 GB is preferable for multiple VMs.
- VM 1: Kali Linux (student security workstation).
- VM 2: Ubuntu Linux or another intentionally configured lab server.
- VM 3: OWASP Juice Shop/WebGoat for web-security exercises.
- Optional: Windows evaluation/lab VM for log-generation exercises.
- Keep vulnerable systems isolated from production networks. Never scan or test systems without explicit authorization.
- Use synthetic usernames, passwords, IP addresses and logs for training wherever possible.

Detailed Module-by-Module Teaching Plan

Module 01: Cybersecurity Fundamentals — 2 Hours

Theory Topics

- What is cybersecurity and information security?
- Security goals: confidentiality, integrity, availability and accountability.
- Assets, threats, vulnerabilities, risks, controls and security incidents.
- Defensive security vs ethical security testing.
- Security roles: user, analyst, administrator, engineer and incident responder.

Hands-On Practical Lab

- Create a simple home/lab asset inventory.
- Classify assets by sensitivity and business importance.
- Map example incidents to asset, threat, vulnerability, risk and control.
- Build a basic personal cybersecurity checklist.

Free / Lab Tools: Kali Linux or Ubuntu, Browser, Text editor

Expected Outcome: Student can explain the basic cybersecurity vocabulary and security lifecycle.

Module 02: Networking & Network Security Basics — 2 Hours

Theory Topics

- OSI and TCP/IP concepts.
- IP addresses, MAC addresses, ports and protocols.

- TCP vs UDP; DNS, HTTP/HTTPS, SSH and ICMP.
- Private vs public addressing and basic subnet concepts.
- Firewalls, segmentation, VPNs, IDS/IPS and secure network design basics.

Hands-On Practical Lab

- Use ipconfig/ifconfig/ip and route commands to inspect a lab machine.
- Use ping and traceroute/tracert in the student's own lab.
- Use Nmap against localhost or an explicitly authorized lab VM.
- Identify open ports and relate them to services.
- Capture a simple DNS/HTTP/TCP exchange in Wireshark.

Free / Lab Tools: Nmap, Wireshark, Kali Linux/Ubuntu

Expected Outcome: Student can read basic network information and identify common services and controls.

Module 03: CIA Triad, Threats & Attack Concepts — 2 Hours

Theory Topics

- Confidentiality, Integrity and Availability.
- Threat actors, attack surface, attack vectors and indicators.
- Malware, phishing, credential attacks, denial of service, web attacks and insider risk.
- Threat vs vulnerability vs risk.
- Security controls: preventive, detective, corrective and compensating.

Hands-On Practical Lab

- Analyze five incident scenarios and identify the affected CIA properties.
- Create a threat-to-control mapping.
- Build a simple attack-surface diagram for a fictional small company.
- Discuss how one attack can affect multiple CIA properties.

Free / Lab Tools: MITRE ATT&CK website (reference), Whiteboard/diagram tool, Synthetic scenarios

Expected Outcome: Student can describe common attack concepts and select appropriate defensive controls.

Module 04: Vulnerability Assessment Fundamentals — 2 Hours

Theory Topics

- Vulnerability, exposure, exploit and risk.
- CVSS concept and severity vs business impact.
- Asset discovery and service enumeration.
- Vulnerability-scanning workflow.
- False positives, validation and remediation priorities.

Hands-On Practical Lab

- Discover hosts/services in an isolated lab.
- Run a safe vulnerability assessment against a deliberately vulnerable local target.
- Review findings and separate informational, low, medium and high-priority issues.
- Document evidence, impact and recommended remediation.
- Perform a basic re-test after remediation.

Free / Lab Tools: Nmap, OWASP Juice Shop/WebGoat, Optional Greenbone/OpenVAS if local resources permit

Expected Outcome: Student can conduct a basic authorized vulnerability-assessment workflow and write findings.

Module 05: Password & Authentication Security — 2 Hours

Theory Topics

- Authentication factors: knowledge, possession, inherence.
- Passwords, passphrases, password managers and MFA.
- Hashing vs encryption.
- Salting and password-storage concepts.
- Brute force, password spraying, credential stuffing and rate limiting.
- Account lockout and secure authentication design.

Hands-On Practical Lab

- Generate strong passphrases and compare estimated attack resistance conceptually.
- Create a local test account and inspect secure authentication settings.
- Use sample hashes in an offline lab to demonstrate hashing and salting concepts.
- Analyze synthetic authentication logs for repeated failures.
- Design a secure authentication policy for a fictional company.

Free / Lab Tools: Linux, Python, Offline sample hashes, Synthetic logs

Expected Outcome: Student can explain authentication risks and recommend practical password/MFA controls.

Module 06: Access Control Fundamentals — 2 Hours

Theory Topics

- Authentication vs authorization vs accounting.
- Least privilege and need-to-know.
- RBAC, ABAC and basic DAC/MAC concepts.
- User, group, file and resource permissions.
- Privilege escalation as a security risk.
- Access reviews and separation of duties.

Hands-On Practical Lab

- Create users and groups in a Linux lab.
- Configure file permissions using chmod/chown/chgrp.
- Test access as different users.
- Identify excessive permissions and remediate them.
- Build a simple role-to-permission matrix.

Free / Lab Tools: Linux, Python optional, Spreadsheet/editor

Expected Outcome: Student can implement and review basic access controls in a lab environment.

Module 07: Web Application Security Basics — 2 Hours

Theory Topics

- How a web application works: browser, server, HTTP request/response.
- URLs, methods, headers, cookies, sessions and status codes.
- Input validation and output encoding.
- Common risks: injection, broken authentication, access-control failures, XSS and insecure configuration.
- Secure development basics and OWASP Top 10 awareness.

Hands-On Practical Lab

- Launch a deliberately vulnerable local web application.
- Inspect requests and responses using browser developer tools and OWASP ZAP.

- Identify a safe, local example of an input-validation issue.
- Observe cookies/session behavior.
- Document the vulnerability, evidence, impact and remediation.
- Repeat the test after applying a defensive fix where feasible.

Free / Lab Tools: OWASP ZAP, OWASP Juice Shop/WebGoat, Browser

Expected Outcome: Student can explain basic web-security risks and perform safe testing on a local vulnerable application.

Module 08: Cryptography Fundamentals — 2 Hours

Theory Topics

- Plaintext, ciphertext, key and algorithm.
- Encoding vs encryption vs hashing.
- Symmetric encryption and key management.
- Asymmetric encryption and public/private keys.
- Digital signatures, certificates and PKI.
- TLS/HTTPS at a conceptual level.
- Why weak or obsolete cryptography is risky.

Hands-On Practical Lab

- Use CyberChef to demonstrate encoding/decoding and hashing.
- Generate a checksum and verify file integrity.
- Create a basic public/private key pair in a local lab.
- Inspect a website's TLS certificate using a browser.
- Compare hashing and encryption using controlled examples.

Free / Lab Tools: CyberChef, OpenSSL, Browser

Expected Outcome: Student can choose the appropriate cryptographic concept for integrity, confidentiality and authentication.

Module 09: Basic Linux & Security Tools — 2 Hours

Theory Topics

- Linux filesystem, users, processes and services.
- Package management and basic security hygiene.
- File permissions and logs.
- Processes, ports and network connections.
- Security-tool workflow and evidence collection.

Hands-On Practical Lab

- Practice pwd, ls, cd, cat, less, grep, find, ps, top, ss, ip and journalctl.
- Inspect authentication/system logs in a lab.
- Find files based on name/permissions.
- Identify listening services.
- Use Nmap for basic service discovery.
- Capture traffic with Wireshark and identify common protocols.

Free / Lab Tools: Kali Linux, Ubuntu, Nmap, Wireshark

Expected Outcome: Student can navigate Linux and use foundational security tools for defensive investigation.

Module 10: Basic Splunk Investigation — 2 Hours

Theory Topics

- What is SIEM and why SOC teams use it.
- Splunk concepts: events, indexes, fields and searches.
- Basic SPL structure.
- Time range, field filtering, stats and sorting.
- Security investigation workflow.
- Alert vs event vs incident.

Hands-On Practical Lab

- Load synthetic Windows/Linux/web authentication logs into an eligible Splunk training environment.
- Search failed and successful logins.
- Filter events by username, source IP and time.
- Use simple SPL commands such as search, stats, table and sort.
- Build a basic failed-login investigation.
- Create a simple dashboard/report if the available edition permits.

Free / Lab Tools: Splunk training/free/trial option, Synthetic logs, SPL

Expected Outcome: Student can perform a basic SIEM search and produce evidence for a simple security investigation.

Module 11: Basic Microsoft Sentinel Investigation — 2 Hours

Theory Topics

- Microsoft Sentinel as a cloud-native SIEM/SOAR platform.
- Log Analytics workspace concept.
- Data connectors, tables and events.
- KQL basics for security analysts.
- Incidents, alerts and entities.
- Basic investigation and hunting workflow.

Hands-On Practical Lab

- Create/use an eligible Microsoft Sentinel free-trial lab where available.
- Onboard a small synthetic or permitted data source.
- Run beginner KQL queries.
- Filter events by time, account, IP or action.
- Review an incident and identify entities/evidence.
- If the cloud trial is unavailable, repeat the KQL/data-analysis exercise using supplied sample data and screenshots.

Free / Lab Tools: Microsoft Sentinel, Log Analytics, KQL, Microsoft Learn sample content

Expected Outcome: Student can navigate Sentinel and perform a beginner-level KQL investigation. Microsoft currently documents a 31-day Sentinel free trial with the first 10 GB/day of Analytics logs free; additional capabilities may incur charges. □cite□turn0search9□

Module 12: AI-Driven Cybersecurity Fundamentals — 2 Hours

Theory Topics

- Generative AI and security operations.
- AI-assisted log summarization, alert triage and investigation.
- Prompting for structured security analysis.

- AI hallucinations, verification and evidence requirements.
- Privacy and safe handling of security data.
- AI for defensive automation and analyst productivity.

Hands-On Practical Lab

- Use a local AI model where possible, or a permitted free AI service with synthetic data.
- Feed a synthetic security alert and ask the model to summarize it.
- Generate an investigation checklist.
- Ask AI to map a scenario to likely MITRE ATT&CK techniques, then manually verify.
- Compare AI output with analyst evidence.
- Create a safe prompt template for SOC analysts.

Free / Lab Tools: Ollama/local model optional, Synthetic logs, Python optional

Expected Outcome: Student understands where AI can assist a security analyst and where human verification is mandatory.

Module 13: Real-World Security Scenarios — 2 Hours

Theory Topics

- SOC-style incident lifecycle.
- Phishing and credential compromise.
- Brute-force login attempts.
- Suspicious outbound traffic.
- Malware/endpoint alert.
- Unauthorized privilege change.
- Web-application attack indicators.

Hands-On Practical Lab

- Scenario 1: investigate repeated login failures.
- Scenario 2: analyze a phishing-style synthetic email.
- Scenario 3: investigate suspicious outbound connections.
- Scenario 4: investigate an endpoint alert using supplied evidence.
- For each scenario: identify facts, hypothesis, evidence, severity and recommended action.
- Create a short incident timeline.

Free / Lab Tools: Splunk or Sentinel where available, Wireshark, Synthetic email/logs, Browser

Expected Outcome: Student can apply multiple fundamentals to a realistic SOC-style investigation.

Module 14: Hands-On Practical Exercises — 1 Hour

Theory Topics

- Evidence collection and documentation.
- Security checklist thinking.
- Basic triage and prioritization.
- Communicating findings clearly.

Hands-On Practical Lab

- Complete a timed mini-lab containing network discovery, log analysis, authentication review and a web-security observation.
- Record commands, evidence, findings and remediation.
- Present a 5-minute analyst briefing.

Free / Lab Tools: All course lab tools

Expected Outcome: Student demonstrates practical application across the course rather than isolated tool usage.

Module 15: Capstone Cybersecurity Projects — 3 Hours

Theory Topics

- Capstone methodology: scope, asset inventory, threats, evidence, findings and remediation.
- Risk-based prioritization.
- Professional security-report structure.
- Lessons learned and security improvement plan.

Hands-On Practical Lab

- CAPSTONE A – Small Company Security Assessment: map assets, identify risks, perform authorized discovery, document findings and propose controls.
- CAPSTONE B – Mini SOC Investigation: use synthetic logs in Splunk/Sentinel or the local alternative to investigate a multi-stage incident.
- CAPSTONE C – Secure Web Application Review: assess a deliberately vulnerable local application and produce a remediation report.
- Final presentation: 10 minutes + questions.
- Submit evidence pack: scope, screenshots, commands/queries, findings, timeline and remediation plan.

Free / Lab Tools: Nmap, Wireshark, OWASP ZAP, Splunk/Sentinel where available, Kali Linux, Synthetic logs

Expected Outcome: Student completes a beginner-level end-to-end cybersecurity project and produces a professional report.

30-Hour Teaching Schedule

Session	Modules	Focus	Hours
1	Modules 01–02	Cybersecurity foundations + networking	4
2	Modules 03–04	CIA/threats + vulnerability assessment	4
3	Modules 05–06	Authentication + access control	4
4	Module 07	Web application security lab	2
5	Modules 08–09	Cryptography + Linux/security tools	4
6	Modules 10–11	Splunk + Microsoft Sentinel investigations	4
7	Modules 12–13	AI security + real-world scenarios	4
8	Module 14	Integrated practical exercise	1
9	Module 15	Capstone projects and presentations	3

Assessment Structure

- Knowledge checks / quizzes – 15%
- Hands-on lab completion – 30%
- Security investigation exercises – 20%
- Capstone project – 25%
- Final presentation / viva – 10%

Student Deliverables

- Personal cybersecurity checklist
- Basic network/security diagram
- Vulnerability-assessment report
- Authentication security policy
- Access-control matrix
- Web-security finding report
- Cryptography/integrity exercise evidence
- Linux security command worksheet
- Splunk investigation worksheet
- Microsoft Sentinel/KQL investigation worksheet
- AI-assisted security-analysis prompt and verification record
- Final capstone report and presentation

Instructor Delivery Method

- Recommended pattern for each 2-hour block: 45–50 minutes theory, 60–65 minutes guided lab, 5–10 minutes recap/quiz.
- Demonstrate first, let students reproduce second, then give a small independent challenge.
- Use screenshots and command/query worksheets for beginners.
- Use only synthetic data or data the instructor/student is authorized to inspect.
- Keep vulnerable machines isolated and never expose intentionally vulnerable services to the public Internet.
- For cloud trials, monitor usage and billing settings before onboarding data. Sentinel's current free-trial documentation explicitly notes that extra capabilities can still incur charges. □cite□turn0search9□

Safety & Ethical Use

All scanning, vulnerability assessment and web-security testing in this course must be performed only against systems that the learner owns or has explicit authorization to test. Use deliberately vulnerable applications and isolated virtual machines. Do not use course exercises to attack public websites, third-party systems, accounts, networks or services. Kali's own documentation warns that misuse of penetration-testing tools without authorization can cause serious damage and consequences. □cite□turn0search11□

Course Positioning

This is an independent professional training program designed for foundational cybersecurity learning. It is not affiliated with, endorsed by, or a certification program of Splunk, Microsoft, Kali Linux, OWASP, or any other referenced vendor/project.

Recommended Next-Level Courses

- L1 Security Operations Center (SOC) Analyst – 60 Hours
- Splunk Admin & Developer – 60 Hours
- Splunk Enterprise Security – 60 Hours
- Microsoft Security Operations & Defender Sentinel Practitioner – 60 Hours
- Certified Ethical Hacking & AI – Expert Course – 120 Hours